

ENTRY 029 — A Provably Correct Fix That Made the Model Worse — August 13, 2026 — Status: COMPLETE (fix not shipped)

Entry 029 — When Being More Right Makes the Model More Wrong

Chasing the Remaining bell_near / bell_far Gap Finds a Real Formula Bug, and a Reason Not to Fix It Yet

1. The Remaining Gap After Entry 028

bell_near_77_82 (10.81-point gap) and bell_far_0_126 (11.21-point gap) were still the two worst circuits after Entry 028's fixes. Both routes dwell on qubit 80, $T_1 = 5.0 \mu\text{s}$ -- the same extreme-decoherence qubit Entry 023 originally flagged. Isolating gate error, decoherence, and readout separately for the single worst leg (the (80,79) SWAP in bell_near's route) against Aer found gate error and readout both matched closely, but the emulator's decoherence-only prediction for that one leg (70.58%) was far below Aer's isolated result (93.15%).

2. A Real, Derivable Bug -- Verified

Every decoherence call site since Entry 023 has computed a hazard probability $\gamma = 1 - e^{-t/T_1}$ and used $(1 - \gamma)$ directly as the success multiplier -- treating amplitude damping as an all-or-nothing failure. Working through the Kraus operators for one qubit of a Bell pair under amplitude damping shows this is wrong: the decay-triggered branch (probability $\gamma/2$) always leaks outside {00,11}, but the no-decay branch (probability $1 - \gamma/2$) ALWAYS still lands inside it. The correct success factor is $1 - \gamma/2$, not $1 - \gamma$. Verified directly: an isolated single-qubit amplitude-damping Aer test at four T_1 values (5, 10, 20, 50 μs) matched $1 - \gamma/2$ to within shot noise at every single point.

3. Applying It Made Things Worse

The fix was implemented and re-validated against the full Entry 022 Kyiv reference set. bell_mid_77_100 and bell_far_0_126 both improved. bell_near_77_82 got dramatically worse -- its gap nearly tripled, from 10.81 to 24.08 points -- because relaxing the decoherence penalty pushed an already-too-optimistic prediction even further from the real answer. On the full 5-circuit set, mean gap moved from 5.13 to 6.61 points and worst-case from 11.21 to 24.08. A four-way sweep (single-carrier vs. both SWAP endpoints charged, times γ vs. $\gamma/2$) confirmed Entry 028's exact combination -- single carrier, full γ -- is the best-performing option of the four on this reference set, despite $\gamma/2$ being the more physically correct formula in isolation.

combination	mean gap	worst gap
single carrier, full γ (Entry 028, shipped)	5.14	11.08
single carrier, $\gamma/2$ (this entry's attempt)	6.66	24.25
both endpoints, full γ	6.45	19.33
both endpoints, $\gamma/2$	5.62	13.23

Entry 029 — Conclusion

The correct single-qubit formula does not survive contact with the full model. A naive check confirms why: multiplying Aer's own individually-isolated gate-error, decoherence, and readout success numbers together for bell_near's route gives roughly 72% -- but the real combined-noise result is 49%. Even a perfect decomposition of each channel in isolation, recombined by simple multiplication, misses a 23-point interaction effect. Entry 028's technically-wrong γ formula was, on this route, accidentally absorbing some of that missing interaction term. Fixing the formula in isolation removed a real bug and simultaneously removed the accidental compensation for a different, larger, unmodeled one -- leaving the combined prediction worse. The $\gamma/2$ fix is NOT shipped. The codebase is unchanged from Entry 028.

4. What This Actually Means

The project's multiplicative model -- gate error, decoherence, and readout as independent probabilities multiplied together -- has a structural limit on routes that dwell on extreme- T_1 qubits. The channels are not independent there: whatever damage decoherence does to the state appears to make it MORE vulnerable to the other error sources than treating the channels separately predicts. Closing this gap for real would require modeling that interaction directly, not

patching the decoherence formula alone -- a materially bigger task than this entry's scope.

5. Next Step

Given the interaction effect is specific to extreme-T1 routes and the current model (Entry 028's state) already gives a 5.13-point mean gap across the reference set -- a large improvement over where the project stood two entries ago -- task #21 (extending the D-screen protocol to Sherbrooke and Cairo) is the better use of the next entry. The channel-independence question is left open, documented here rather than chased further, and can be revisited if a future chip's reference set turns up the same pattern.

Research Log — Index Addendum (Entry 029)

Entry	Title	Date	Status
Entry 029	A Correct Formula Fix That Regressed the Combined Model	Aug 13, 2026	Complete (not shipped)

t1_survival() ($1 - \gamma/2$) is derived and verified correct in isolation but NOT applied in `emulator_v4.py` or `exact_dwell_routing.py` -- it regresses the Entry 022 Kyiv reference set. Codebase left at Entry 028's state. Documented as an open channel-independence question for future chips.